

Networking Deep Dive

BGP • VPN/IPsec • Transit Gateway • Architecture & Troubleshooting
Week 3 Study Summary | Internal Training Documentation

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1. Introduction

This document captures the learning outcomes from Week 3 of the six-month Network DevOps internship programme at Autodesk, which is focused on networking fundamentals and advanced cloud connectivity. The Week 3 deliverable required a summary write-up covering three core topics: Border Gateway Protocol (BGP), VPN/IPsec tunnelling, and AWS Transit Gateway (TGW). The week also included shadowing a live troubleshooting call and walking through at least one real routing issue with the mentor.

Building on the AWS fundamentals studied in Weeks 1 and 2 — including VPC architecture, subnets, route tables, Security Groups, and NACLs — this week's content steps into the more advanced territory of dynamic routing protocols, encrypted tunnelling, and multi-VPC connectivity at enterprise scale.

The content is informed by two live learning sessions with mentor Bharathraj B R, covering the Autodesk corporate AWS Cloud WAN architecture, as well as independent study using AWS documentation and technical guides.

2. BGP — Border Gateway Protocol

2.1 What Is BGP?

Border Gateway Protocol (BGP) is the routing protocol that underpins the global internet. It is an Exterior Gateway Protocol (EGP), meaning it is designed to exchange routing information between different autonomous systems (ASes) rather than within a single network. BGP is sometimes referred to as the “protocol of the internet” because virtually all internet traffic is ultimately routed using BGP decisions.

In cloud networking, BGP is used primarily to enable dynamic route exchange between an organisation's on-premises infrastructure and cloud environments such as AWS. Rather than manually maintaining static route tables, BGP allows network devices to advertise and learn routes automatically, and to respond to topology changes without human intervention.

2.2 BGP Key Concepts

The following concepts are foundational to understanding how BGP operates:

Concept	Explanation
Autonomous System (AS)	A collection of IP networks under the control of a single organisation or operator, identified by a unique Autonomous System Number (ASN).
ASN (AS Number)	A 16-bit or 32-bit number assigned to each autonomous system. Private ASNs (64512–65534) are commonly used in hybrid cloud VPN configurations.
BGP Peer / Neighbour	Two BGP routers that have established a BGP session between each other and exchange routing information.
Prefix / Route Advertisement	A BGP speaker advertises a network prefix (e.g., 10.0.0.0/24) to its peers, informing them that it can reach that network.
BGP Attributes	Properties attached to each route that influence path selection. Key attributes include: AS_PATH (the list of ASes a route has traversed), LOCAL_PREF (preference within an AS), MED (metric advertised to peers), and NEXT_HOP.
EBGP vs. IBGP	External BGP (EBGP) runs between different autonomous systems (e.g., on-premises and AWS). Internal BGP (IBGP) runs within the same AS.
MD5 Authentication	BGP sessions can be protected with MD5 authentication. Both peers must be configured with the same password; a mismatch prevents the session from establishing.
Route Filtering	BGP filters allow administrators to control which prefixes are accepted from or advertised to specific peers, preventing unwanted route propagation.

💡 How BGP Path Selection Works

BGP selects the best path to a destination using a defined sequence of attributes:

1. Highest LOCAL_PREF (preferred within the local AS)
2. Shortest AS_PATH (fewer hops through other autonomous systems)
3. Lowest MED (metric advertised by neighbouring ASes)
4. Prefer EBGP over IBGP routes
5. Lowest router ID (tiebreaker of last resort)

Understanding this sequence is critical for influencing traffic flow in hybrid cloud environments.

2.3 BGP in AWS: Site-to-Site VPN Dynamic Routing

AWS Site-to-Site VPN supports two routing modes: static and dynamic. When dynamic routing is selected, BGP is used to exchange routes between AWS and the on-premises customer gateway device. This allows the network to adapt automatically when routes change or when a tunnel fails, without requiring manual configuration updates.

In the AWS Transit Gateway + VPN configuration, each VPN connection consists of two IPsec tunnels for redundancy. BGP sessions are established over each tunnel, and routes are exchanged dynamically. ECMP (Equal-Cost Multi-Path) routing across both tunnels is supported when the VPN is attached to a Transit Gateway, enabling higher aggregate throughput and true active-active failover.

With BGP dynamic routing, administrators can also influence traffic paths by adjusting BGP attributes such as AS_PATH prepending and MED values in their BGP advertisements.

2.4 Static vs. Dynamic (BGP) Routing

Aspect	Static Routing	Dynamic Routing (BGP)
Configuration	Routes manually defined	Routes exchanged automatically
Failover	Manual intervention required if tunnel fails	Automatic failover when tunnel is lost
Scalability	Limited — each new network requires a new static entry	Scales to thousands of routes without manual changes
Use Case	Simple topologies, single VPN, no redundancy	Enterprise hybrid cloud, multi-region, high availability
Best Practice	Acceptable for dev/test environments	Recommended for all production hybrid cloud connections

3. VPN and IPsec

3.1 What Is a VPN?

A Virtual Private Network (VPN) is a technology that creates a secure, encrypted communication channel over an untrusted network such as the public internet. VPNs are used to extend a private network across geographic boundaries, allowing remote sites, users, or cloud environments to communicate as if they were on the same local network.

In cloud networking, Site-to-Site VPN is the most relevant type. It connects an on-premises data center or office network to a cloud network (such as an AWS VPC) over an encrypted IPsec tunnel, enabling hybrid connectivity.

3.2 IPsec Fundamentals

IP Security (IPsec) is a protocol suite that provides authentication and encryption for IP packets. It is the standard underlying technology for most enterprise VPN implementations, including AWS Site-to-Site VPN. IPsec operates at Layer 3 of the OSI model and protects traffic between defined endpoints.

IPsec operates in two main phases:

IPsec Phase	Purpose
Phase 1 — IKE Negotiation	Establishes a secure control channel (ISAKMP SA) between the two peers. The peers authenticate each other and negotiate encryption and hashing algorithms. This uses the Internet Key Exchange (IKE) protocol.
Phase 2 — IPsec SA	Establishes the data channel (IPsec SA) over which actual user traffic is encrypted and transmitted. Uses the parameters negotiated in Phase 1 to protect the data tunnel.

IPsec operates in two modes:

- Tunnel Mode — the entire original IP packet is encrypted and encapsulated inside a new IP packet. This is the mode used for Site-to-Site VPN between network gateways.
- Transport Mode — only the IP payload (data) is encrypted, while the original IP header remains. This is typically used for end-to-end communication between individual hosts.

IPsec Tunnel Parameters to Know

IKE Version: IKEv1 or IKEv2 (AWS supports both; IKEv2 preferred for new deployments)

Encryption Algorithm: AES-128, AES-256 (symmetric encryption for the data channel)

Integrity / Hashing: SHA-1, SHA-256 (for packet authentication)

Diffie-Hellman Group: DH Group 2, 14, 19, 20 (for key exchange security)

Pre-Shared Key (PSK): A shared secret used to authenticate IPsec peers

Dead Peer Detection (DPD): Detects when a peer becomes unreachable and initiates failover

3.3 AWS Site-to-Site VPN

AWS Site-to-Site VPN is a fully managed service that enables encrypted IPsec connections between on-premises networks and AWS. Each VPN connection consists of two tunnels for redundancy. AWS terminates the VPN on a Virtual Private Gateway (VGW) or a Transit Gateway (TGW) attachment.

Key components of an AWS Site-to-Site VPN:

- Customer Gateway (CGW) — an AWS resource that represents the on-premises VPN device. It is configured with the device's public IP address and BGP ASN.
- Virtual Private Gateway (VGW) or Transit Gateway Attachment — the AWS-side VPN endpoint. TGW is preferred for connecting multiple VPCs.
- VPN Connection — the logical connection between the CGW and VGW/TGW, comprising two IPsec tunnels with separate outside IP addresses for redundancy.
- Tunnel Configuration — each tunnel has its own pre-shared key, inside IP addresses (for BGP peering), and IKE/IPsec parameters.

3.4 VPN + BGP: Dynamic Hybrid Connectivity

Combining AWS Site-to-Site VPN with BGP dynamic routing is the recommended approach for production hybrid cloud connectivity. In this architecture, BGP sessions are established inside each IPsec tunnel. The on-premises router and the AWS gateway exchange route advertisements, automatically propagating network prefixes in both directions. [\[6\]](#)

Both the IPsec tunnel and the BGP session must be terminated on the same customer gateway device. This means the device must be capable of running both IPsec and BGP simultaneously — most enterprise-grade routers and firewalls support this requirement.

In the Autodesk environment, the network also includes IPsec site-to-site VPN attachments within the Cloud WAN core network, alongside Direct Connect Gateway attachments and Transit Gateway peerings, providing multiple redundant paths for hybrid connectivity.

4. AWS Transit Gateway (TGW)

4.1 What Is Transit Gateway?

AWS Transit Gateway (TGW) is a regional, fully managed network transit hub that enables centralised connectivity between VPCs, on-premises networks, and other cloud resources. Rather than creating a mesh of individual VPC peering connections, a Transit Gateway acts as a central router to which all networks attach. [\[4\]](#)

Transit Gateway significantly simplifies network management at enterprise scale. Without TGW, connecting N VPCs would require $N \times (N-1) / 2$ individual peering connections and complex route table management. With TGW, each VPC attaches to the central hub, and routing policies are managed centrally.

Transit Gateway is a regional construct: each TGW serves traffic within a single AWS Region. For cross-region connectivity, TGWs in different regions can be linked using TGW Peering, and multiple TGWs can be connected to AWS Cloud WAN to form a global network.

4.2 TGW Core Components

The following components form the building blocks of a Transit Gateway architecture. These were discussed in detail during the live Cloud WAN learning sessions.

Component	Description
TGW Attachment	A logical connection between a resource and the Transit Gateway. Supported attachment types include: VPC, Site-to-Site VPN, Direct Connect Gateway, TGW Peering, and Connect (SD-WAN).
TGW Route Table	A set of routes that determine how traffic is forwarded between attachments. Each attachment can be associated with one route table, and a single TGW can have multiple route tables for traffic segmentation.
TGW Peering	A connection between two Transit Gateways in the same or different AWS Regions, enabling inter-region connectivity. Peering connects TGWs themselves, not individual VPCs.
Propagation	Routes from an attachment can be automatically propagated into a TGW route table, reducing manual configuration overhead.
ASN	Each Transit Gateway is assigned a private BGP ASN. This ASN is used when establishing BGP sessions with on-premises devices via Site-to-Site VPN attachments.

💡 TGW Route Tables in Practice

During the Cloud WAN session, Bharathraj explained that in the Autodesk environment, Transit Gateway route tables are attached to Cloud WAN and used to define traffic paths between VPCs and on-premises networks.

Each attachment is assigned to a route table, and routes can be either statically defined or dynamically propagated via BGP.

Multiple route tables allow different segments of the network (e.g., prod, dev) to have different routing policies without impacting each other.

4.3 TGW vs. VPC Peering

Aspect	VPC Peering	Transit Gateway
Architecture	Point-to-point (mesh)	Hub-and-spoke (star)
Scalability	Difficult to scale — $N \times (N-1)/2$ connections required	Scales easily — all VPCs attach to one hub
Routing	Separate route table entry per peering	Centralised route tables on the TGW
Cross-Region	Supported via inter-region VPC peering	Supported via TGW Peering or Cloud WAN
Bandwidth	No inherent limit per connection	Up to 50 Gbps per TGW attachment
Cost	Lower for small topologies	More cost-effective at enterprise scale

5. TGW Architecture — Autodesk Corporate Network

5.1 From Transit Gateway to AWS Cloud WAN

Autodesk's corporate AWS network previously used Transit Gateways as the primary mechanism for interconnecting VPCs and on-premises resources across regions. Over time, as the network grew to thousands of VPCs and four AWS regions, the team migrated to AWS Cloud WAN for centralised management.

The migration was driven by the need for a single console view of all attachments and the ability to enforce network-wide policies centrally. While Transit Gateways remain part of the architecture (peered into the Cloud WAN core network), the Cloud WAN global network now serves as the overarching control and routing plane.

5.2 Cloud WAN Architecture Summary

Autodesk operates a global AWS Cloud WAN network with the following high-level architecture. This was reviewed during the AWS Account Access learning session and the AWS Cloud WAN Session 2.

Architecture Element	Details
Global Networks	Two global networks: one for testing/sandbox and one for production. The production network is the primary environment for all corporate workloads.
AWS Regions	Four approved regions: US East 1 (N. Virginia), US West 2 (Oregon), AP Southeast 1 (Singapore), EU West 1 (Dublin). Corporate resources may only be deployed in these four regions.
Core Network Edges (CNEs)	Each region hosts a Core Network Edge, which functions as a regional router within the Cloud WAN. All CNEs are fully interconnected, enabling any-to-any route advertisement and traffic forwarding across regions.
Total Attachments	Approximately 1,554 attachments in the production network. The majority (~95%) are VPC attachments. The remainder includes Direct Connect Gateways, Transit Gateway route tables, and IPsec site-to-site VPN attachments.
Direct Connect Gateways	One Direct Connect Gateway per region (four total), providing dedicated private connectivity from on-premises data centres to the Cloud WAN core network.

5.3 Attachment Types and Counts

The table below summarises the types of attachments observed in the Autodesk Cloud WAN production environment:

Attachment Type	Approximate Count	Notes
VPC Attachments	~1,480+	The vast majority of attachments. Each customer account typically has one VPC, though some accounts have multiple.
Direct Connect Gateway	4 (one per region)	Provides dedicated, private, high-bandwidth connectivity from on-premises data centres.
TGW Route Table Attachments	Multiple	Legacy Transit Gateway route tables peered into the Cloud WAN for continued connectivity during and after migration.
IPsec Site-to-Site VPN	Multiple	Encrypted tunnels for connecting external or partner networks to the Cloud WAN.

5.4 Segmentation and Policy-Based Routing

One of the most powerful features of Cloud WAN is its ability to enforce network segmentation through segments and policies. In the Autodesk environment, VPCs are grouped into segments based on their environment type (e.g., dev, stage, prod). [\[2\]](#)

- **Segments** — Logical groupings of VPCs and attachments. VPCs within the same segment can communicate with each other. Cross-segment communication is blocked by default.
- **Automatic Segment Assignment** — VPCs are not manually assigned to segments. Instead, environment tags (e.g., environment: prod) trigger automated scripts that attach the VPC to the correct segment automatically.

- Policies — Explicit policies must be created to allow cross-segment traffic. For example, allowing prod to communicate with a shared-services segment requires a network sharing policy to be defined.
- Palo Alto Firewalls (Network Function Groups) — When cross-segment communication is permitted, traffic is routed through Palo Alto firewalls deployed as Network Function Groups within the Cloud WAN. This enforces security inspection before allowing data to traverse segment boundaries.
- Policy Rule Numbering — Each policy rule is assigned a sequence number. Rules are evaluated from lowest to highest, and the first matching rule is applied — the same behaviour as access lists and route maps in traditional networking.

E Architecture Key Insight

The move from Transit Gateway to Cloud WAN enabled Autodesk to manage a network of 3,000+ VPCs across four regions from a single console.

The previous Transit Gateway architecture required managing multiple regional TGWs, their peering connections, and route tables independently.

Cloud WAN centralises all of this — policies, segments, attachments, and routing — into a single global network configuration.

6. Troubleshooting: Shadowing a Live Call

6.1 Troubleshooting Framework

A structured troubleshooting methodology is essential in complex cloud networking environments where issues can span multiple layers — from physical connectivity and IPsec negotiation to BGP session state and route propagation. The following framework was observed and discussed during the live shadowing session:

Step	Action	Description
1	Identify the Symptom	Understand exactly what is failing: is connectivity completely lost, intermittent, or one-directional? Identify affected source and destination.
2	Check Attachment Status	In the AWS console, verify that all relevant TGW or Cloud WAN attachments are in the 'Available' state. Any error flags indicate an attachment-level issue.
3	Review the Routing Information Base	Examine the Cloud WAN or TGW Routing Information Base (RIB) to determine what routes are being learned and which are being installed as best routes.
4	Check BGP Session State	If BGP is in use, verify that BGP sessions are Established on both peers. A session in Idle or Active state indicates a connectivity or configuration problem.
5	Validate IPsec Tunnel State	For VPN-connected networks, verify that both IPsec tunnel phases (IKE Phase 1 and Phase 2) are up. A tunnel in 'IPSEC IS DOWN' state will prevent BGP from establishing.
6	Check Route Table Propagation	Confirm that routes from the relevant attachment are being propagated into the correct TGW or Cloud WAN route table.
7	Review Security Group / NACL Rules	Once routing is confirmed correct, verify that Security Groups and NACLs on the destination resource allow the relevant traffic on the correct port and protocol.
8	Escalate with Evidence	If the issue remains unresolved, capture screenshots and log data and escalate to the relevant team (e.g., cloud management or on-premises networking team) with a clear problem statement.

6.2 Real Routing Issue Walkthrough

The following routing issue was walked through during the Week 3 shadowing session as a practical example of applying the troubleshooting framework above.

Issue: VPN Connection Could Not Be Established

Observed Symptom: VPN connect button was unresponsive; the VPN tunnel did not come up.

Initial Assessment: Piyush raised the issue during the AWS Account Access session. The VPN client was running but the connection button was not responding.

Action Taken: Bharathraj advised capturing a screenshot of the exact error state and sharing it with the support team for investigation.

Lesson Learned: Even simple VPN access issues can block broader learning and should be escalated promptly with clear evidence. Screenshots of error states are essential for efficient support triage.

Outcome: Ticket raised with the VPN support team for investigation.

A second routing concept explored during the session was the behaviour of duplicate routes in the Cloud WAN Routing Information Base:

- When two attachments advertise the same route prefix (e.g., 10.0.0.8/32), only the best route is installed in the routing table.

- Route selection follows standard routing device behaviour: the best route is determined by metrics such as AS_PATH length and route origin.
- The non-best routes remain in the RIB as backup paths and are promoted if the best path becomes unavailable.
- This behaviour is identical to traditional router routing tables and ensures efficient, loop-free forwarding even in large-scale networks.

7. Key Takeaways

The following learning outcomes were achieved through Week 3 of the internship programme:

#	Learning Outcome
1	Understood BGP as the internet's core routing protocol and its role in dynamic route exchange between autonomous systems.
2	Learned the key BGP attributes (AS_PATH, LOCAL_PREF, MED, NEXT_HOP) and how they influence path selection in hybrid cloud environments.
3	Understood the two phases of IPsec (IKE negotiation and IPsec SA) and the parameters required for tunnel establishment.
4	Learned how AWS Site-to-Site VPN uses two redundant IPsec tunnels, and how BGP runs inside these tunnels for dynamic routing.
5	Understood Transit Gateway as a regional hub-and-spoke routing architecture and its core components: attachments, route tables, propagation, and peering.
6	Understood the difference between static and dynamic (BGP) routing and why dynamic routing is preferred for production hybrid cloud environments.
7	Gained hands-on knowledge of the Autodesk Cloud WAN architecture: four regions, ~1,554 attachments, segment-based routing, policy enforcement, and Palo Alto firewall inspection.
8	Understood how Autodesk migrated from Transit Gateway to AWS Cloud WAN to enable centralised, global network management at scale.
9	Learned a structured eight-step troubleshooting framework applicable to cloud networking issues spanning IPsec, BGP, routing, and security controls.
10	Developed practical documentation skills by capturing and structuring learning outcomes in a format suitable for management review.